

A Bayesian Approach to Ulnar Collateral Ligament (UCL) injury prediction - Project Proposal

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Abstract

Ulnar Collateral Ligament (UCL) reconstruction, commonly referred to as Tommy John Surgery, has seen a significant rise among Major League Baseball (MLB) pitchers, prompting growing interest in identifying the mechanical and performance-based factors that contribute to injury risk. While previous studies have examined these relationships using traditional frequentist approaches separately, this study combines multiple different model techniques to present a broad framework for finding significant predictors of UCL Surgery. These models include Lasso and Ridge Regression, Principal Component Regression (PCR), Partial Least Squares Regression (PLS), Random Forest, Multiple Linear Regression, and a Bayesian Statistical Model. Using these models, our findings aim to refine injury prediction models and provide a more comprehensive statistical framework for understanding the mechanics underlying UCL injuries in professional baseball.

1 Introduction

1.1 Overarching Problem/Objective

Pitchers get injured, it's a problem that is faced in all levels of baseball. From those who play in high school, in college, and make it to the MLB everyone will face a injury some point in their career. Even with all this, Tommy John Surgery (or UCL Reconstruction) has seen a significant rise in occurrence across the MLB. According to surveyed

statistics, 25% of Major League Baseball pitchers and 15% of Minor League Baseball pitchers having undergone a UCL surgery. [3] This project has the goal of digging into why this is happening, and trying to use Bayesian Statistical methods of approach to find possible factors of injury, and do better then ML/Frequentist methods at predicting possible factors of injury for future MLB players.

1.2 Literature Context and Background

A lot of research has been done to predict injuries in other sports, and predicting baseball injuries using frequentist/ML methods. To name a few, T-Testing has been used on multiple occasions [6] [2], trend testing with a Mann-Kendall test, [4], goodness of fit tests via a Chi-Squared [6], and naive Bayes with a support vector machine [7]. All of these studies have given a variety of mixed results, with some finding that while pitch velocity does not have a direct risk factor of UCL injury, pitch type can have a possible significance of risk of injury, [2] and that there is statistical significance between the throwing of a fastball and possible UCL tearing. [1]

2 Resources and Methods

2.1 Data Collection/Acquisition

All data collection has been planned and done already. Thanks to the baseballR package [5], I have had no problem collecting the data needed for this

project. Data collection was split into two portions, pitches who did/did not have UCL surgery during their career in the MLB. Data were collected from 2018 to 2024, with pitches collected 2-years prior to injury occurring.

2.2 Planned Tasks and Analysis

To facilitate the building and use of a Bayesian Hierarchical Model, multiple ML/Frequentist models will be built first to better understand my data, and what variables have possible correlation with injury significance. Some of these models include:

- Multiple Linear Regression - To get a baseline understanding of my variables and which should be selected for other models. This, along with Weighted Multiple Linear regression, will serve to weed out variables that have no correlation present.
- Weighted Multiple Linear Regression - See Multiple Linear Regression
- Lasso Regression - A better approach than Multiple Linear Regression, this model will work to "shrink" my predictors and give feedback on which ones seem to have the largest "influence" over predicting injury.
- Random Forest - The goal of using Random Forest is to see the weights that are given to each variable in the building of trees. Multiple "stumps" will be built first, followed by one very deep tree to understand prediction.
- PCR and PLS - Used when variables have high correlation between each other, this model works well to search through the "weeds" of my data and relate individual columns of information to prediction rather than getting stuck relying on other columns of data to build strong predictors.

2.3 Timeline of Work

- Middle of February - I plan to have over half of my ML/Frequentist models done. Since all the programming is done in R, I hope to build out a

well-documented quarto document where I hold all significant information.

- End of February - Complete ML/Frequentist models and documentation.
- Mid-March/End of March (Depending on Time) - Begin the building of a hierarchical model. I hope to have completed some basic trial runs of the model by the time April 1st hits. I also hope to have written more of my paper by this point, having the methodology completed.
- SPARK Day - By the time I present at SPARK, I will have preliminary results completed and evidence (hopefully) to support my findings. This will be presented both at SPARK and the Sports Analytics Symposium at UCONN.
- Graduation - By my date of graduation, I will have completed my paper and prepared it for submission to a journal for review and possible publishing.

2.4 How is this practical?

This research project is practical because technically at this point I am half-way done with it. I have done a lot of work already preparing my models and I have some built from prior work I did last year. Along with this, I have methods I can fall-back upon (IE Survival Analysis instead of a Bayesian Model) in case I do fail. This project is something I've been wanting to complete as well, and that point is what drives me to finish it by the time I graduate.

References

- [1] Steven F DeFroda, Peter K Kriz, Amber M Hall, David Zurakowski, and Paul D Fadale. Risk stratification for ulnar collateral ligament injury in major league baseball players: a retrospective study from 2007 to 2014. *Orthopaedic journal of sports medicine*, 4(2):2325967115627126, 2016.
- [2] Robert A. Keller, Nathan E. Marshall, John-Michael Guest, Kelechi R. Okoroha, Edward K.

- Jung, and Vasilios Moutzouros. Major league baseball pitch velocity and pitch type associated with risk of ulnar collateral ligament injury. *Journal of Shoulder and Elbow Surgery*, 25(4):671–675, 2016.
- [3] David P. Leland, Stan Conte, Nicole Flynn, and et al. Prevalence of medial ulnar collateral ligament surgery in 6135 current professional baseball players: A 2018 update. *Orthopaedic Journal of Sports Medicine*, 7(9), 2019.
- [4] Benjamin C. Mayo, Adam Miller, Michael J. Patetta, Garrett R. Schwarzman, Jeffrey W. Chen, Marshall Haden, Erwin Secretov, and Mark R. Hutchinson. Preventing tommy john surgery: The identification of trends in pitch selection, velocity, and spin rate before ulnar collateral ligament reconstruction in major league baseball pitchers. *Orthopaedic Journal of Sports Medicine*, 9(6):23259671211012364, 2021. PMID: 34189147.
- [5] Bill Petti and Saiem Gilani. *baseballr: Acquiring and Analyzing Baseball Data*, 2024. R package version 1.6.0, <https://github.com/BillPetti/baseballr>.
- [6] Daniel A Portney, Lucas T Buchler, Jake M Lazaroff, Stephen M Gryzlo, and Matthew D Saltzman. Influence of pitching release location on ulnar collateral ligament reconstruction risk among major league baseball pitchers. *Orthopaedic journal of sports medicine*, 7(2):2325967119826540, 2019.
- [7] David Whiteside, Douglas N Martini, Adam S Lepley, Ronald F Zernicke, and Grant C Goulet. Predictors of ulnar collateral ligament reconstruction in major league baseball pitchers. *The American journal of sports medicine*, 44(9):2202–2209, 2016.